

Both fully compressed tuple questions in arXiv:1809.05154 were answered in arXiv:2005.11582

Literature-resolution packet for arXiv:1809.05154

12 August 2026

Status: literature already answered. Davidson and Passer give a positive answer to uniqueness of fully compressed tuples and a necessary-and-sufficient criterion for existence of a fully compressed presentation. Their paper explicitly says that these results answer the questions of Passer and Shalit.

1 Source questions

For a tuple $T \in \mathcal{B}(H)^d$, its matrix range is

$$\mathcal{W}_n(T) = \{\phi(T) : \phi : \mathcal{B}(H) \rightarrow M_n \text{ is unital completely positive}\}, \quad \mathcal{W}(T) = \bigcup_{n \geq 1} \mathcal{W}_n(T).$$

The tuple is *fully compressed* when compression to every proper closed subspace strictly decreases its matrix range.

On page 15 of the official arXiv PDF of arXiv:1809.05154, Questions 4.8 and 4.9 ask:

For tuples $A, B \in \mathcal{B}(H)^d$ which are not necessarily compact, if A and B are fully compressed and $\mathcal{W}(A) = \mathcal{W}(B)$, does it follow that A is unitarily equivalent to B ?

Let $\mathcal{C}$ be a closed and bounded matrix convex set. When can $\mathcal{C}$ be written as the matrix range of a fully compressed tuple $T \in \mathcal{B}(H)^d$?

2 Exact later answers

Davidson–Passer, arXiv:2005.11582, proves both results and cites the source paper directly.

Uniqueness

Corollary 2.13 on page 11 states that if S and T are fully compressed d -tuples and $\mathcal{W}(S) = \mathcal{W}(T)$, then S and T are unitarily equivalent. Immediately before it, the authors write that this gives a positive answer to Passer–Shalit Question 4.8. The proof uses equality of matrix ranges to obtain a complete order isomorphism of the associated separable operator systems, then Corollary 2.12 upgrades that isomorphism to a unitary equivalence.

Thus the answer to the first source question is **yes**, with no compactness hypothesis.

Existence

Associate to a closed bounded matrix convex set $\mathcal{C}$ its separable abstract operator system $\mathcal{S}_{\mathcal{C}}$ (equivalently, use any tuple T with $\mathcal{W}(T) = \mathcal{C}$; every such $\mathcal{C}$ has a tuple realization). Corollary 3.6 on page 15 proves the equivalence

$$\begin{aligned} \mathcal{S}_{\mathcal{C}} \text{ has a fully compressed concrete presentation} &\iff \\ \text{the isolated points of } \widehat{C_e^*(\mathcal{S}_{\mathcal{C}})} &\text{ are dense.} \end{aligned}$$

Equivalently, $\mathcal{S}_{\mathcal{C}}$ must admit a concrete presentation completely normed by irreducible representations that are strongly peaking. This is the necessary-and-sufficient answer to the second source question. The later paper explicitly says it is answering the Passer–Shalit question, though it cites it as Question 4.19 rather than Question 4.9 in the arXiv PDF.

The same paper also supplies intrinsic specializations. For example, if $\mathcal{C}$ is the closed matrix convex hull of its absolute extreme points, then Corollary 4.7 says that such a fully compressed tuple exists exactly when $\mathcal{C}$ is the closed matrix convex hull of its crucial matrix extreme points.

3 Status, scope, and audit

The match is explicit and complete: uniqueness is Corollary 2.13, and the general existence criterion is Corollary 3.6. These are later-literature answers, not new results of this run. No part of either source question is left open within the separable setting relevant here; Davidson–Passer also show that every fully compressed tuple acts on a finite-dimensional or separable Hilbert space.

Search audit. The run’s cheap indexes had no prior record for arXiv:1809.05154. Exact-phrases and title searches of arXiv found arXiv:2005.11582. The official source and PDF of both papers were inspected; the decisive passages are source page 15 and supporting pages 11 and 15.

Human review. Recommended as a high-confidence literature match. Verify only the harmless 4.9/4.19 numbering discrepancy and the standard passage from a closed bounded matrix convex set to its abstract operator system.

References

- [1] B. Passer and O. M. Shalit, *Compressions of compact tuples*, arXiv:1809.05154; Linear Algebra Appl. 564 (2019), 264–283.
- [2] K. R. Davidson and B. Passer, *Strongly peaking representations and compressions of operator systems*, arXiv:2005.11582; Int. Math. Res. Not. IMRN 7 (2022), 5037–5070.